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Social Force as a Microscopic Simulation Model for Pedestrian Behavior in Crowd Evacuation

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Understanding the evacuation mechanism has been an important subject nowadays in order to model and simulate pedestrian behavior in crowd evacuation for the case study of the emergency scenario such as building on fire. Almost two decades, the interest of Social Force Model (SFM) in pedestrian traffic increased significantly. SFM is a microscopic model and has been proposed as a realistic simulation model and preferred choice which imitate the pedestrian movement. The social interactions among the pedestrians, environment, and collision are emergent phenomena and very hard to capture, rare and like the evasive maneuvers in order to avoid dangers in crowd evacuation. This research focused on SFM simulation model for a case study in terms of crowd dynamic, pedestrian dynamic and evacuation dynamic. This shows that SFM is one of the accurate and realistic model to simulate the pedestrian behavior in crowd evacuation

Keywords— Crowd Evacuation, Social Force Model, Fire Emergency Scenario, Evacuation, Microscopic Simulation

1. INTRODUCTION

T Crowd evacuation occurs when an emergency happens and it involves the agents. The agents considered are the human or also known as the pedestrian. Hence the pedestrian behavior faces the emergency case in the crowd evacuation and this case study focus on fire emergency scenario in the building which leads to crowd and the pedestrians involve in evacuation to egress. It aims to devise a simulation model to illustrate situations that are almost same as real case (case study) or so called as a realistic model. The SFM Model is one of model which is widely discoursed in the world of Crowd Dynamic.

This is the most popular technique has been used by the researchers introduced by [1] and will be discussed details in the next section. This research work provided the fundamental and the concepts of crowd dynamic together with pedestrian dynamic. It is presented the microscopic model which is use Social Force Model as the proposed approach. The case study of this research work covers the panic scenario of the pedestrian in fire emergency scenario for the crowd evacuation. The aim of the evacuation is the safety and avoids injury and the goal of a simulation model is effective and realistic as real case. The criteria of SFM have been studied and then it is proposed as good simulation model to present the case studies as discussed above. All the elements of SFM also described in details and related to the case study and the issues to be settled. The research questions to be answered in this study are as follow; what is the criteria and significance elements to investigate and modeling the crowd evacuation? The second one is how to have a realistic model of the emergence phenomena in crowd evacuation using Social Force Model (SFM).

The organization of this research work is depicted in Figure 1, this research work focus on Social Force Model for the case study of crowd evacuation fire emergency scenario. In the next section is the related works and the existing literature; about the microscopic model, crowd dynamic, pedestrian dynamic and the crowd evacuation for fire emergency scenario The next section is Social Force Model (SFM), continue with social-force approach. Then, the next chapter described

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the SFM in the case study, the methodology, research outcome and the las one is conclusion with the future works. The case study already described before with the scope of pedestrian behavior for crowd evacuation (the characteristic) lead the researchers to work on the SFM (popular and well-known approach) as a simulation model [2].

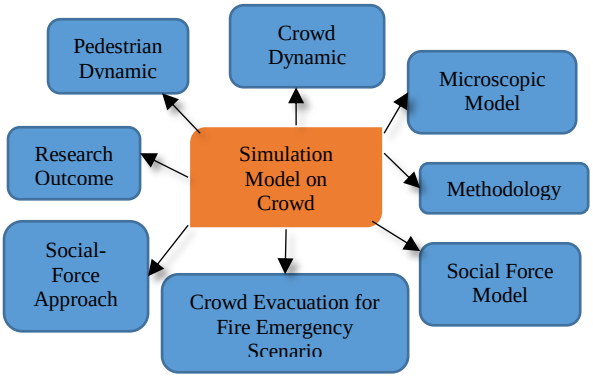


Fig.1. Research Work Organization

2. RELATED WORKS

This section described briefly the existing works about the simulation model; microscopic model with Social Force approach so called as Social Force Model (SFM). The microscopic model also presented the pedestrian dynamic, crowd dynamic, evacuation dynamic and social interaction for the crowd evacuation relates to pedestrian behavior.

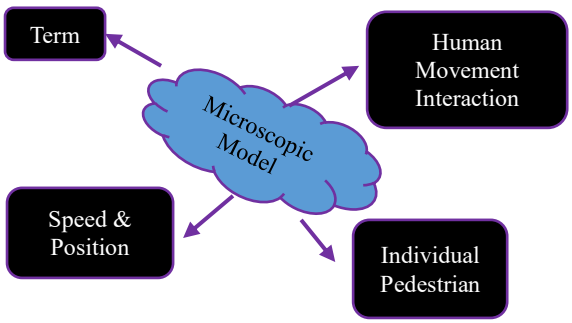


Fig.2. the foundation of Microscopic Model

Based on [3], the perspective of crowd evacuation and the simulation model has been discussed. Simulation models able to ease to understand the predict one and make comparison with the current system through the testing scenario using specific software tools. Figure 3 presented the fundamental of crowd evacuation. Initially, the pedestrian dynamic divided to grid-based and social-based approach [1,4] This is a great theoretical significance for the emergency evacuation design [5] In the category of pedestrian dynamic, there are crowd dynamic, it is focused on the pedestrian movement such as walking in a group which able to imitate it realistically [5].

Normally social force models as the basic model to simulating the crowd dynamic. The pedestrian movement also known as the multi-modal transportation system in crowd dynamic [6]. The goals are to have an efficient, comfortable and safe evacuation for the pedestrian (based on the behavior). This element is one part of evacuation dynamic; study about the behavior of pedestrian to evacuate from crowd for the safety

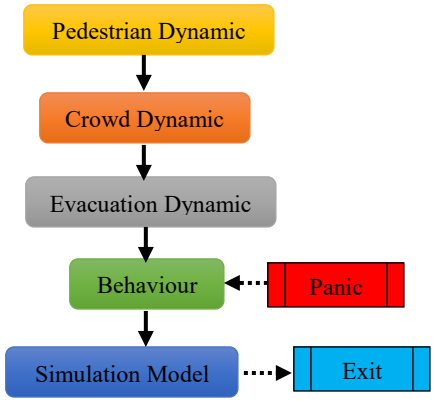


Fig.3. the fundamental concept of crowd evacuation simulation model

purposes with the same goal for egress to exit(s). In order to achieve the goal, it should have a proper evacuation system to avoid injuries and save many lives [7,8]. This is lead to social force elements which the familiarity is part of evacuation behaviors [7]. Familiarity is relates to the knowledge of the route or environment and this behavior is compliance to real time information and has a great impact in the evacuation system [7].

Modelling and empirical study of pedestrian behavior is imperative to analyze and assess safety precautions for those situations [6]. Social force model is most suitable approach to model and simulate the crowd evacuation in normal and emergency scenario [9]. The factor so called as force-based for the human behavior in the crowd evacuation is the route choice which is desire for the quickest route and to be with others [26]. Other than that, the force-based approach; relates to the shortest time for each pedestrian goal which is egress at the same time in the same place. The factor of real time rerouting also important and significance to be considered [26] in this research work.

The social interaction in the group behavior is one of main factor, and focused on the important element such as the individual pedestrian movement in crowd for the same aims and goals considered the forming interactions with others and affecting each other's trajectories [10] . Social Force Model (SFM) as a microscopic model for pedestrian behavior in crowd evacuation relates to panicking and blocking situation because of crowd evacuation process for any emergency cases may lead to serious and critical injuries and the worst case is death [11,12].

However, during the emergency, the emergent and some characteristic tends to the factor of physical and psychological. This is capture the pedestrian behavior of egress to the exit(s) during this situation will be observed

because the pedestrian trample and ram for their own safety. These elements contribute to force-based [11,13] approach in order to simulate the crowd evacuation.

The next section will discuss about the concept of social force model and the application for the case study of fire emergency scenario in the building for capture the pedestrian behavior in crowd evacuation.

3. CONCEPT OF SOCIAL FORCE MODEL

Social Force Model is a well-structured [9] and well-suited which accurate for the prediction and simulation flow performance [9]. It consist the human movement interaction by individually in detail [4,14] as discussed before it involve the speed and position and refer to [15] described that the common model for the dynamic of pedestrian crowds is Social Force Model. In SFM, the rules is well-known in the crowd dynamic area which is the equation and formulae [2,16]. This research work not describe in details about the equation of each element in SFM; able to refer to SFM approach introduced by Dirk Helbing. In SFM, each pedestrian movement referred to desired speed and desired direction [5 17.]

SFM closed to simulation scheme which described these for steps as below in order to capture the pedestrian movement and behavior;

- Starting point is a state (the position) and determine the velocity and acceleration
- Desired speed for each pedestrian specify if the density is reducing, the desired speed contrast with the time
- Desired point of each pedestrian is the trajectory of end point
- The motion of pedestrian is described as β, trajectory as λ and initial actual location as $r_\alpha(t)$

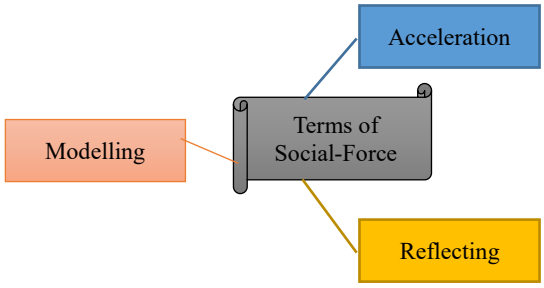


Fig.4. Social-Force for Pedestrian

In the figure 4 is presents the terms in social force for pedestrian in acceleration, reflecting and modelling. Acceleration described the desired velocity of motion (main parameter in the SFM to measure the performance) [1]. The term of reflecting refers to the pedestrian to keep certain distance between the pedestrian each other and the distance between the pedestrian and borders. The next term is modelling which is referring to attractive effects of each movement for the pedestrian [1,18,19]. Social Force Model (SFM) with the basic rules of human behavior have characteristic of well predictable, automatic reaction, the pedestrian motion and with the standard solutions [30].

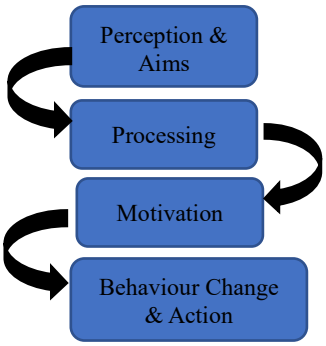


Fig.5. Social-Force concept for pedestrian

The SFM concept for pedestrian was depicted in figure 5 which described the perception and aims. Then, will processing each movement and the effect from motivation (force-based) for other pedestrian and then the behavior of pedestrian change and action will be taken by the pedestrian since crowd evacuation using SFM [30]. When walking in complex built environment, pedestrians' actions are governed by both goal-directed and exploratory behaviors (Gibson, 1988). Goal-directed behavior typically refers to the motivation of moving to certain points in space such as residential buildings and transit terminals. Exploratory behavior is usually stimulated by visually attractive objects along the path to the goal, for instance, window displays and street performance [20].

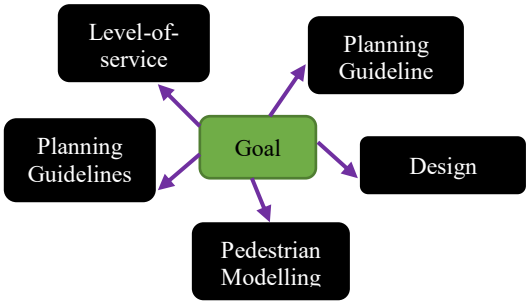


Fig.6. Social-Force Goal for Pedestrian

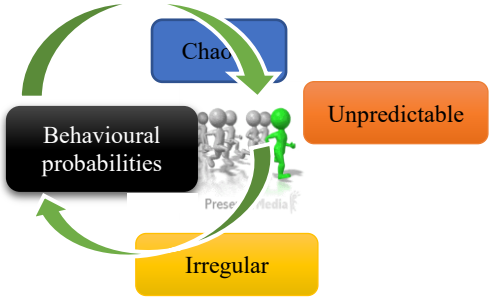


Fig.7. Pedestrian characteristic

Figure 6 presented the goal of SFM for the pedestrian in order to have a realistic simulation model for a rare case like the fire emergency scenario of a crowd evacuation. The goals are very important and significance as stated before; the human movement as a complicated and complex task to predict the flow and the behavior, and it only can be simulated with the intelligent tool to have a realistic simulation model (as real case).

The important element in this case study and research work is the actor or so called as pedestrian. The emergent scene happens for the phenomenon affected the motion of the pedestrian in panic situation. Escape panics of the pedestrian caused of unexpected and dangerous scenario such as the fire emergency and leas to crowd evacuation. The process of evacuation from crowd to the same goal of egress and need to be analyzed the clogging and bottleneck [21] scene using SFM as a microscopic model in the pedestrian dynamic and it is called "faster-is-slower" effect [21,22.]

Figure 7 depicted the characteristic of pedestrian in crowd and in the panic situation [1] when fire emergency scenario is happen in the building . Normally the real behavior of each pedestrian in panic scene included the element of chaotic, unpredictable, irregular and behavior probabilities [1,10]. The bottleneck with clogging or arching condition happen leads to high density and may affecting the departure flow and the arrival flow increase. The SFM aim to move continuously but it will impact the movement if some pedestrians are impatient and start pushing. This case is triggers the phantom panic and it is hard to simulate it in order to have a realistic model.

4. SOCIAL-FORCE APPROACH FOR CROWD EVACUATION SIMULATION MODEL

The aim for the crowd evacuation is safety and avoids injury to save many lives and this is accepted as a proper evacuation. Therefore, better preparations of evacuation guidance are significant. The details to simulate and modelling the crowd evacuation related to the cased study is based on the movement is change based on the collective phenomena in panic: -

1. Pedestrian nervous and higher fluctuation
2. Try escape from source of panic and increase desire velocity
3. In complex situation, don't know how to do right things and tend to do what others do (social based factor)

In order to model the situation as above, this research work proposed to use SFM approach. In the last two decade, [15] Social Force Model already used and has been introduced by Helbing and the application and interest to use it in pedestrian dynamic (human movement) increased as well. It is significance to choose as proposed model to simulate the crowd evacuation [23] in order to observed the pedestrian behavior as in the real crowds. Social force is incurred the movement and actions by the pedestrian and the characteristic of SFM is continuous and not discrete in time and space. The mechanism and flow of evacuation has been significance to understand it [7] due to emergent phenomena occur in crowd and it is rare [24]. The SFM approach needs to follow the movement scheme to be applied for the simulation model. It has two types of movement which are the desired speed and desired direction [5] based on the rules and algorithm and the schemes as follow:-

- The starting point which is the state (position) and the parameters of velocity and acceleration to be considered
- Desired speed of each pedestrian; the density is decrease, the desired speed constant with the time
- The desired point of each pedestrian to the end point of trajectory (goal)
- Motion of pedestrian as β, trajectory of pedestrian as α, time period as λ, as $r_\alpha(t)$ with the relative distance error (simulation versus tracked)

Based on [16] , one of the research outcome is SFM Formulae; by the most popular and was the introducer of SFM. The formulae are as follow and can be used in this research work due to use SFM as a model for the crowd evacuation simulation: -

i. Certain destination from initial to the goals: desired area at every time for nearest point to corresponding area (one of factor for real crowd evacuation to apply SFM as the model for this research work)
ii. Motion of pedestrian is influenced by other pedestrian which the density and desired speed (comparison the desired and actual movement speed)
iii. The pedestrian will feel uncomfortable to the strange person who react in aggressive way (repulsive effect of other pedestrian)
iv. Pedestrian requires the space for the next step
v. Pedestrian with certain distance of borders, walls, streets, obstacles in order to avoid danger or injury (repulsive effect)
vi. The pedestrian attracted by the other person such as friends (as an active agent)

Other than that, the most popular topic discussed in Social Force Model is the effect of ''faster-is-slower''. This is meant if the velocity is increased, so that the movement in a large crowd is decreased or the traffic flow becomes slower [25.] The emergent scenario can be seen for the big crowd and be a serious emergency case if the desired velocity is high. This is the elements of the slower-is-faster effect on crowd and its evacuation. This situation will produce the negative effect and the low

density with high velocity impact the evacuation process and may take longer time. The experimental result from [4] shows that in the low density scenario, the pedestrian will act in bad and unexpected manner such as become not rational as normal person because of the big crowd and everyone want to egress from it.

The pedestrian has been observed and then the pedestrian mostly moving around repeatedly somewhat to find the route safely egress to the exit. This is already discuss before which is the pedestrians move to the desired point with or without to avoid the collision either the other pedestrians or obstacles [4]. The emergency scenario while crowd happen lead to panic situation cause the behavior of pedestrian become abnormal instead of to find the shortest or fastest route to the desired point [25]. This is conducted to clogging and bottleneck will be happening and the pedestrian keep heading forward to the desired point to egress from the crowd.

If the density is increased, the high velocity and the speed of evacuation become slower; the competition between all pedestrians becomes high in a small space in order to find the way to egress. This may lead to bodily interaction and friction slow down the total crowd motion. [25]. From this effect of faster-is-slower as described, this work proposed to have three models for simulate and modeling the crowd evacuation (the same case study) with SFM, Agent-based Simulation (ABS) and the hybrid of SFM/ABS.

5. RESEARCH METHODOLOGY

Research methodology is a theory about the methods and the details process involved of a research work [26]. Many agencies faced the issues with the modeling and simulation of any cases such as crowd evacuation especially in emergency scenario for example the physicists, computer scientists[27] and etc. and it is become of tremendous attention as well. One of the important process is extracted the real data or experiments and this is the same process will be used for this research work. The main processes should have in a simulation model for emergency crowd evacuation such as the experimental tools and then the test-case scenario and results as proposed by [26]. This research work also uses the combinatorial of pedestrian and environment modeling; fire scenario and the exit(s) as egress is the evacuation goal.

The method of a simulation model [28] able to apply for this research work in order to have an effective model has been presented in figure 8. From the process of case study and data collection, a research theory came out to interpret with a model and can be implemented using the simulation to validate and verify the simulation model based on the case study.

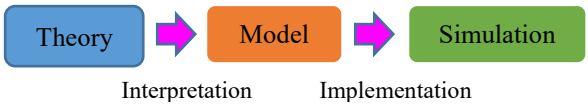


Fig.8. the method for a simulation model

Figure 9 depicted the methodology for this research work for the crowd evacuation based on the case study [29]. The case study already identified and then it is involved the process of data collection. The primary data collect from an organization via an interview session [30]. The secondary data collected from literature of existing work. The conceptual model develops and the start the development phase and implementation process with the experiment of the case studies include the process of experimental observation of crowd evacuation [31] . The experimental results allow the researchers to continue the phase of verification and validation. The comparison and analysis of the experimental results will be done in the final phase of this work.

The implementation for develop a simulation tool to apply the social force model (SFM) for crowd evacuation for the case of fire emergency in the building, can be used any appropriate modelers, toolkits, or programming language. Normally any programming languages are appropriate such as Java, C, C#, and etc. The modelers include the library, integrated development environment (IDE) and hybrid application as well. For the toolkit include the all-purpose software and programming language as the language already stated before and the tools like Anylogic, Promodel, Netlogo, Anylogic can be used. In this work we will use the Anylogic Software with Java Programming Language.

Based on [32], the element of force-based in the crowd dynamic plays the important role, therefore the concept of continuous force-based model as follow; the relationship between the agent and agent and also the agent versus obstacles. The work includes the collision avoidance calculation as one of the method and it will apply for this research work. Other than that, the method such as calculate the evacuation time is the most significance one [28,33]. The goals of evacuation (theory) has been discuss before which is include these elements; avoid obstacle and moving towards the exit or egress from the crowd.

6. RESEACH OUTCOME & DISCUSSION

Referred to the case study as described before, it is to settle the issues of pedestrian behavior and the movement in the crowd evacuation which is emergent scenario in emergency and cannot be simulated in real.

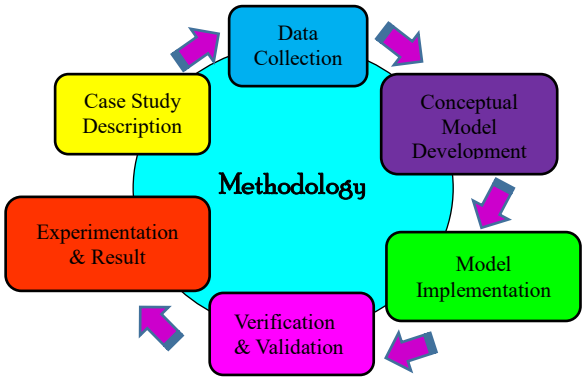


Fig.9. Research methodology for simulation modelling

It can be validated n verified such as by the fire drill training and to compare the experimental result with the analysis of real case training result. The simulation model of SFM as the proposed technique in order to have an enhance simulation model based on this case study for an effective and realistic one. The contribution in the research work is the fundamental concept, criteria and the significance elements for crowd evacuation simulation model of fire emergency scenario. It is described in terms of the pedestrian dynamic, crowd dynamic to the evacuation dynamic and included the behavior with panic scenario (as depicted in figure 3).

The research methodology as one of the contribution for this study which is depicted in Figure 9 is applying in order to have an effective simulation model for the case study of building on fire. It is about the pedestrian movement in crowd evacuation; to have as realistic model as real scenario and crowd evacuation is impractical in cost and safety to verify the real crowds (emergency). From this research work, the researchers contributed the fully and details of crowd evacuation and the approach of Social Force Model. The researchers also proposed to apply SFM in simulation model for the case study in order to achieve the goals and objective.

7. CONCLUSIONS

The insights of the research study are related to movement dynamic which is come from the crowd and its evacuation and based on the pedestrian behavior lead the researcher proposed SFM as a model of social-based approach to cater the issues as discussed. In conclusion, SFM is capable to describe self-organizing of observed of collective of pedestrian behavior realistically as a good simulation model for fire emergency scenario in crowd evacuation. This simulation platform can be used by researchers to implement human-like behaviors. The concept, rules, formulae, and the flow of Social Force Model already discussed and can be applied for the implementation of a simulation model for crowd evacuation (fire emergency scenario) [24].

The method can be applied for research work in order to have an effective and realistic simulation model for crowd evacuation of fire emergency scenario as the case study is divided into three phase; theory and interpret it using a model and can be implemented by a simulation to predict the movement of pedestrian. Therefore, the researcher came out with six phases of methodology in order to implementing the solution of the issues and applies SFM in experimental simulation. For the future works, the proposed model will be used to simulate the fire emergency scenario in crowd evacuation in order to capture the emergent pedestrian behavior with panic scene as realistic as real life case. Hence, the research work with the hybrid model has been proposed to analyze and compare the results between three models; SFM, Agent-based Simulation (ABS) and the hybrid of SFM/ ABS with the method of experiment and analysis of the experimental results. The parameter, factors and specific scenario which impact the results as the measurement performance for the same case studies can be reviewed [34, 35].

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